15.6.1 Description

React is a popular JavaScript library for building user interfaces, particularly single-page applications. It allows developers to create reusable UI components and manage the state of their applications efficiently. React is often used in conjunction with other libraries and frameworks to build full-stack applications.

In this hands-on lab, we will build a full stack web application using Laravel 12 and React. The backend will expose a RESTful API for managing `Product` data, and the frontend will be a React interface for performing Create, Read, Update, and Delete operations. We will use the Laravel Starter Kit with React and SQLite as the database.

15.6.2 Objectives

By the end of this lab, you will be able to:

- Scaffold a Laravel 12 project with the React starter kit using Laravel CLI
- Create a RESTful Product API using Laravel
- Connect the React frontend to the backend via `axios`
- Perform full CRUD operations from the React interface
- Use SQLite as the persistent database

15.6.3 Prerequisites

Before starting, make sure the following are installed:

- PHP ≥ 8.2
- Composer
- Laravel Installer (`composer global require laravel/installer`)
- Node.js and npm
- SQLite installed on your system
- A text editor (e.g., Visual Studio Code)

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

15.6.4 Steps